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INTERNATIONAL INSTITUTE
OF MEDICAL SCIENCE &
TECHNOLOGY COUNCIL

MAJOR PROJECT REPORT
on
**IoT-Driven Cold Chain System for Real
Time Monitoring and Predictive Failure
Prevention**

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ABSTRACT

This project presents an IoT-based cold chain monitoring and control system designed to ensure safe storage of temperature-sensitive products through continuous real-time supervision. Conventional storage methods often rely on manual monitoring, which can lead to delayed response and increased risk of spoilage. The proposed system overcomes these limitations by integrating temperature and humidity sensing with automated control and communication mechanisms. The DHT22 sensor continuously monitors environmental conditions, while the ESP32 microcontroller processes the data and compares it with predefined threshold limits. Based on this analysis, the system activates the cooling mechanism using a Peltier module to maintain the required temperature range.

The system also incorporates a magnetic reed switch to detect door status and enhance monitoring reliability. Real-time data is displayed locally on an LCD and remotely through a mobile application and web dashboard, enabling users to track system performance from any location. In addition, a GSM module is used to send instant alert notifications during abnormal conditions, ensuring timely intervention even in the absence of internet connectivity. The integration of buzzer and LED indicators further strengthens the alert mechanism by providing immediate local notifications.

By combining sensing, automation, and communication, the system improves monitoring accuracy, reduces dependency on manual supervision, and minimizes the risk of product loss. The proposed solution is cost-effective, scalable, and suitable for applications in healthcare, food storage, and pharmaceutical cold chain management.

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CHAPTER 1

PREAMBLE

1.1 INTRODUCTION:

Cold chain systems are essential for maintaining the quality and safety of temperature -sensitive products like vaccines, biologics, and perishable foods. According to the World Health Organization, vaccines need to be stored within a temperature range of 2°C to 8°C to preserve their effectiveness. If these temperature limits are not maintained, the potency of the vaccines can be compromised, which may result in reduced efficacy and pose public health risks.

Ensuring this temperature range during storage and transport is particularly challenging in areas with limited infrastructure. Research indicates that common causes of cold chain failures include equipment breakdowns, power outages, and the absence of continuous monitoring. Conventional cold storage systems often depend on manual checks, which can delay the detection of temperature fluctuations and increase the risk of spoilage.

To overcome these issues, recent advancements in the Internet of Things (IoT) have made it possible to develop intelligent monitoring systems that offer real-time data collection and remote supervision. IoT-based solutions use sensors, microcontrollers, and cloud services to continuously monitor environmental factors such as temperature and humidity. These technologies enhance visibility, maintain compliance with safety regulations, and reduce reliance on manual inspections.

The proposed IoT-based cold chain system leverages an ESP32 microcontroller combined with sensors like the DHT22 to monitor conditions continuously. Data is transmitted to cloud platforms for visualization and analysis, while instant alerts are sent through GSM communication when irregularities occur.

In addition to real-time monitoring, the system utilizes data logging and analytical methods to support predictive maintenance by identifying early signs of potential failures. This forward-looking strategy helps prevent temperature deviations before they happen, minimizing product loss and increasing system reliability.

The goal of this project is to create an affordable and scalable cold chain monitoring solution suitable for practical deployment. By integrating IoT technology with automated control features, the system improves operational efficiency, safeguards product quality, and contributes to the development of smarter logistics and healthcare infrastructure.

1.2 OBJECTIVE:

The main goal of this project is to develop an IoT-based Cold Chain System that guarantees secure storage and transportation of temperature-sensitive products by enabling real-time monitoring and predictive failure detection. The detailed objectives include:

1. To design an efficient cold storage monitoring system

The system will maintain the optimal storage environment (2°C to 8°C for vaccines, in line with WHO recommendations) by integrating sensors and control components to ensure stable operation.

2. To implement real-time monitoring using sensors

Temperature and humidity sensors (such as DHT22) will be used to collect real-time environmental data inside the storage unit. This eliminates reliance on manual checking and ensures continuous supervision.

3. To develop an automated control mechanism

A microcontroller-based system (ESP32) will process sensor data and automatically trigger actions such as activating cooling systems, alarms, or relays when parameters exceed safe thresholds.

4. To enable instant alert and notification system

The system will use GSM or internet-based communication to send real-time alerts (SMS/notifications) to users when abnormal conditions such as temperature rise or door opening are detected.

5. To integrate cloud-based data logging and visualization

The system will store data on cloud platforms, enabling remote monitoring through dashboards. This improves traceability, record-keeping, and compliance with safety standards.

1.3 PROBLEM STATEMENT:

Conventional cold chain systems depend largely on manual checks and periodic monitoring, making them unreliable and prone to human error. Without continuous tracking, temperature variations caused by equipment faults, power interruptions, or improper handling may go unnoticed, leading to deviations from the required storage conditions for different products. For instance, vaccines must be maintained within a strict temperature range of 2°C to 8°C as specified by the World Health Organization, while other perishable goods require different controlled environments. Fluctuations in temperature and humidity, along with frequent door openings, can compromise the quality and safety of sensitive products such as vaccines and food items. While some modern solutions provide basic monitoring capabilities, they often lack real-time alert mechanisms and fail to respond promptly to abnormal conditions. Furthermore, these systems rarely utilize collected data for predictive analysis, resulting in a reactive approach where issues are detected only after damage has occurred. This leads to increased spoilage, economic losses, and potential health risks, highlighting the need for an intelligent, automated, and IoT-enabled cold chain system that ensures continuous monitoring, timely intervention, and improved reliability.

1.4 LITERATURE REVIEW:

[1] The study titled “**Enhancing Food Safety in the Cold Chain Through Internet of Things and Artificial Intelligence**” by Wagner Augusto Müller et al. in the Journal of Food Science (2026) analyzes the impact of temperature variations on food quality and shelf life. The authors emphasize the role of IoT sensors in continuous monitoring and AI algorithms in predicting quality degradation. The system enables data-driven decision-making by identifying patterns in storage conditions. However, the study highlights limitations such as high computational requirements, increased system complexity, and limited real-world implementation. These challenges indicate the need for simpler, cost-effective solutions that can be deployed in practical environments.

[2] The paper “**Development of Disposable Paper-Based Smart Visual Labels with Thermochromic Materials for Cold Chain Temperature Monitoring**” by Qiwen Chen

et al. in ScienceDirect (2025) presents a low-cost monitoring approach using thermochromic materials that change color when temperature thresholds are exceeded. This method provides a simple and intuitive way to detect temperature breaches without requiring electronic components. However, the system is limited by its single-use nature and lack of real-time monitoring capabilities. It also does not provide alerts or remote access, making it unsuitable for continuous cold chain tracking.

[3] The research “**Predicting ‘Broken Chain’ Risks in Food and Pharmaceutical Cold Chains Based on Multi-Modal Data Fusion and Machine Learning**” by Gang Ping (2025) proposes advanced predictive models using IoT data combined with machine learning techniques such as LSTM and Transformer networks. The approach enables early detection of temperature deviations and potential failures. While the model demonstrates high prediction accuracy, it depends heavily on large datasets and computational resources. Additionally, challenges in real-time implementation and system scalability limit its practical applicability.

[4] The paper “**Development of an IoT-Based Smart Cold Storage with Inventory Monitoring System**” by Suganya Agappan et al. (2024) discusses integrating environmental monitoring with inventory management systems. The proposed system improves efficiency by tracking both storage conditions and stock levels simultaneously. It highlights the benefits of combining IoT-based sensing with supply chain management. However, the study lacks detailed implementation of automated control mechanisms and requires further validation under real-world conditions.

[5] The study “**IoT-Based Integrated Sensing and Logging Solution for Cold Chain Management**” (IEEE Xplore, 2025) presents a comprehensive framework for monitoring temperature-sensitive goods using IoT sensors and cloud-based storage. The system ensures continuous data logging and enhances traceability across the supply chain. It supports compliance with safety regulations by maintaining accurate historical records. However, challenges such as sensor calibration issues, data synchronization delays, and infrastructure costs affect system reliability and scalability.

CHAPTER 2

SYSTEM DESIGN

2.1 METHODOLOGY:

The proposed system operates by continuously collecting environmental data such as temperature, humidity, and door status using sensors integrated within the storage unit. The sensed data is processed by the ESP32 microcontroller and compared with predefined threshold values (for example, 2°C to 8°C for vaccines as specified by the World Health Organization). If any deviation from the safe range is detected, the system automatically triggers control actions such as activating cooling mechanisms through relays, generating buzzer alerts, and sending notifications via GSM or internet-based services. The processed data is simultaneously transmitted to a cloud platform (Firebase) using Wi-Fi, where it is stored and visualized through a web or mobile dashboard. This enables users to monitor system conditions in real time and take necessary actions remotely. The system is tested under different conditions to ensure accurate sensing, timely alert generation, and reliable performance.

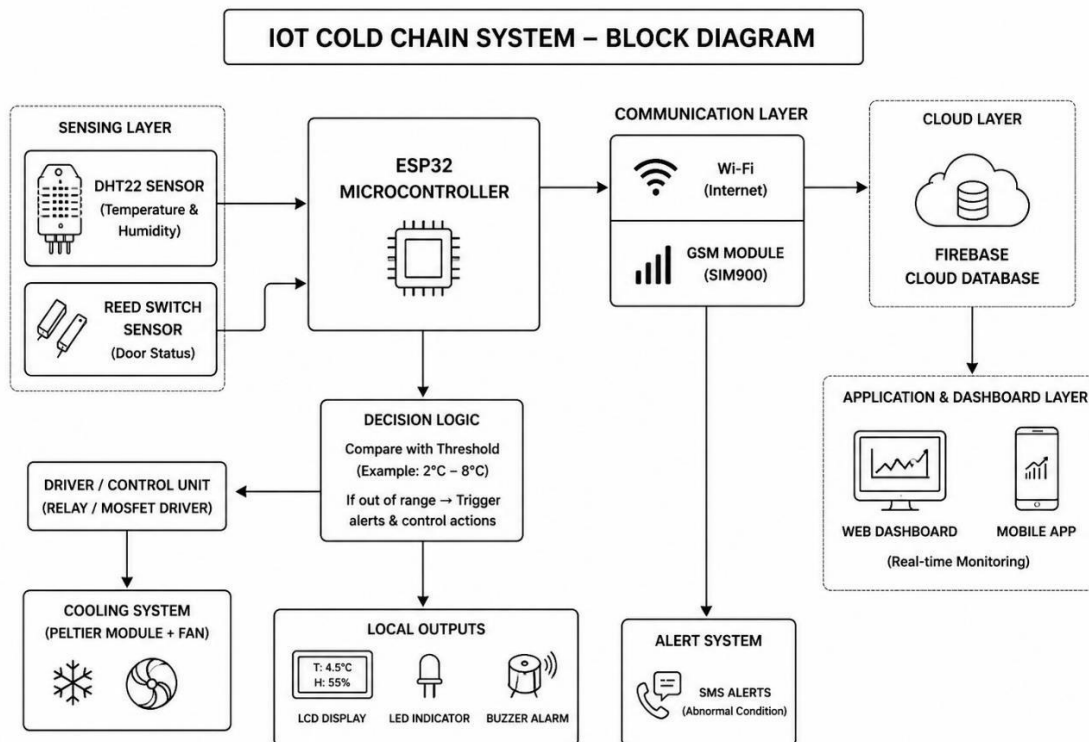


Fig 2.1.1 Block Diagram

CHAPTER 3

HARDWARE USED

3.1 SYSTEM ARCHITECTURE:

The system architecture of the IoT-Driven Cold Chain System is designed to continuously monitor environmental conditions, maintain optimal storage temperature, provide real-time alerts, and enable remote monitoring through cloud integration.

- **Sensing Layer :**

This layer includes sensors such as the DHT22 for temperature and humidity measurement and a magnetic reed switch module for detecting door status. These sensors continuously collect environmental data from the storage unit.

- **Processing Layer :**

The ESP32 microcontroller acts as the central processing unit. It reads sensor data, performs basic processing, and converts it into a structured format suitable for transmission.

- **Decision Layer :**

The system compares the processed data with predefined threshold values (for example, 2°C–8°C for vaccines as specified by the World Health Organization). Any deviation from the safe range is identified as an abnormal condition.

- **Communication Layer :**

The ESP32 transmits data to the cloud using Wi-Fi. Additionally, a GSM module is used to send SMS alerts during critical conditions when immediate attention is required.

- **Cloud Layer(Firebase) :**

Firebase is used as the cloud database to store both real-time and historical sensor data. It ensures reliable data storage, synchronization, and remote accessibility.

• Application & Dashboard Layer :

The system includes a web-based dashboard (developed using the NxtWave platform) and a mobile application interface. These platforms provide real-time visualization of parameters such as temperature, humidity, door status, and system alerts, enabling users to monitor system performance remotely.

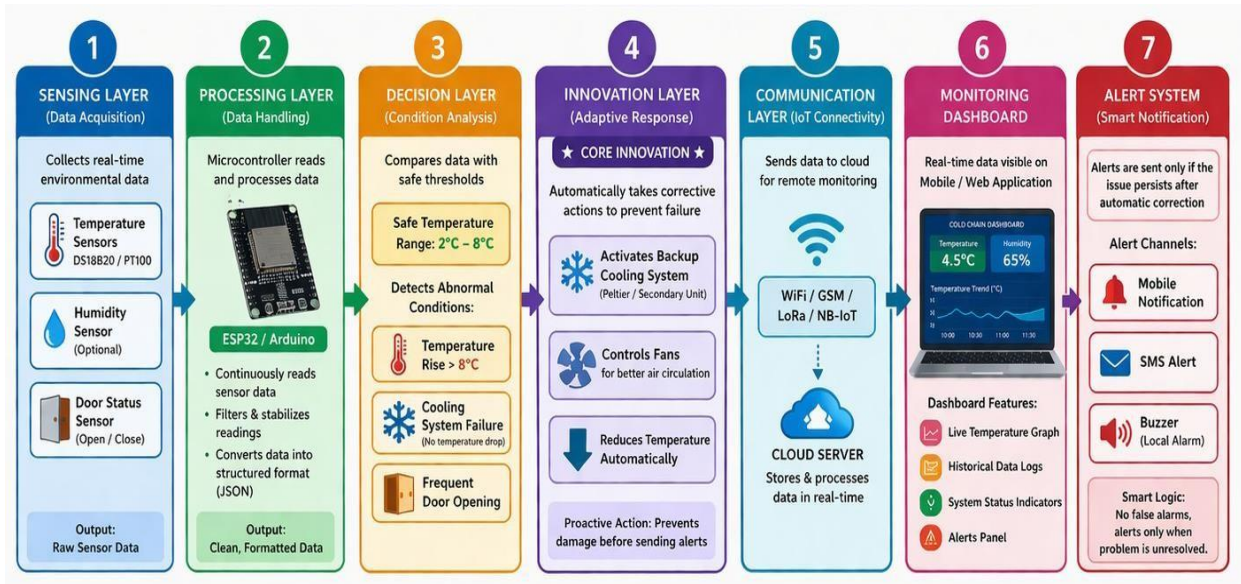


Fig.3.1.1 System Architecture

3.2 HARDWARE COMPONENTS

- ESP32 Development Board
- DHT22 Temperature and Humidity Sensor
- Magnetic Reed Sensor
- 16×2 LCD Display with I2C Interface
- LED Indicator with 220Ω Resistor
- Active Buzzer
- 5V Relay Module
- Peltier Module with Cooling Fan and Heat Sink
- SIM900A GSM Module
- LM2596 Buck Converter

- 12V Power Supply (Adapter/Battery)
- Breadboard and Jumper Wires
- SIM Card

3.3 COMPONENT DESCRIPTION:

3.3.1 ESP32

The ESP32 serves as the central processing unit of the system. It acquires sensor data, performs preliminary processing, and transmits the data to the cloud platform via Wi-Fi.

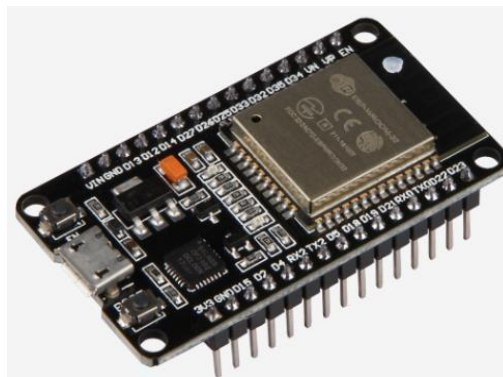


Fig.3.3.1 ESP32

Features of ESP32:

1. Built-in Wi-Fi Connectivity:

The ESP32 supports 2.4 GHz Wi-Fi, allowing it to easily connect to local networks and send real-time data to mobile apps, cloud servers, or IoT platforms.

2. Bluetooth and BLE Support:

In addition to Wi-Fi, the module supports Bluetooth Classic and Bluetooth Low Energy (BLE), providing multiple options for communication and device pairing.

3. Dual-Core Processor:

The ESP32 operates with a dual-core Tensilica LX6 CPU, providing fast performance and enabling simultaneous handling of sensor data, network tasks, and control signals.

4. Low Power Consumption:

Designed with multiple sleep modes, the ESP32 is energy-efficient and suitable for solar-powered systems like this dryer.

5. Rich Peripheral Support:

It includes GPIO pins, ADC, DAC, SPI, I2C, UART, PWM, and capacitive touch sensors, making it extremely flexible for hardware interfacing.

6. Enhanced Security Features:

Supports encrypted communication, secure boot, and flash encryption, ensuring safe data transmission.

Pin Configuration:

Although actual pin layout varies across development boards, the ESP32 commonly includes:

- **GPIO0–GPIO39:** General purpose input/output pins
- **3.3V Pin:** Main power input
- **GND Pin:** Ground reference
- **TX/RX Pins:** UART communication pins for serial data transfer.
- **EN Pin:** Enable/Reset pin
- **SPI Pins (MISO, MOSI, SCK):** For sensor and module interfacing
- **I2C Pins (SDA, SCL):** For communicating with digital sensors

These pins allow the ESP32 to interact with other components like Arduino, DHT22 sensor, and LCD modules.

Working Principle:

- The ESP32 receives humidity and temperature readings from the Arduino Uno.
- It processes and formats the data for wireless transmission.
- Using its Wi-Fi module, it connects to the user's local network.
- It sends real-time environmental data (temperature/humidity) to the mobile application.
- Farmers can remotely monitor drying conditions through the app from any location.
- This makes the system smart, user-friendly, and efficient.
- Smart agriculture and Home automation.
- IoT-based monitoring systems and Wireless sensor networks
- Robotics and automation projects and Environmental monitoring

3.3.2 DHT22 SENSOR

The DHT22 sensor is employed for accurate measurement of ambient temperature and relative humidity within the cold storage environment. It provides digital output, facilitating easy interfacing with the microcontroller.

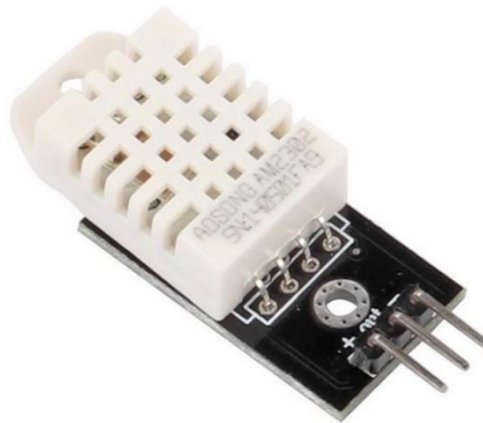


Fig.3.3.2 DHT22 Sensor

Features of DHT22 Sensor:

1. Measures Temperature:

- Range: -40°C to 80°C
- High accuracy suitable for cold storage applications

2. Measures Humidity:

- Range: 0% to 100% RH
- Provides reliable moisture monitoring for sensitive goods

3. Digital Output:

- Sends precise digital signals directly to the ESP32

4. High Accuracy and Stability:

- Better performance compared to DHT11, ideal for critical environments

5. Low Power Consumption:

- Suitable for continuous operation in IoT-based systems

Working Principle of DHT22 Sensor:

The DHT22 uses a capacitive humidity sensing element and an NTC thermistor to measure environmental conditions. Changes in humidity affect the capacitance, while temperature variations change the resistance of the thermistor. An internal IC processes these changes and converts them into calibrated digital signals, which are transmitted to the ESP32 for further processing, monitoring, and control.

3.3.3 MAGNETIC REED SENSOR:

The magnetic reed sensor is utilized to detect the open or closed state of the storage door. This enables monitoring of access frequency, which directly impacts temperature stability.

The sensor is interfaced with the ESP32, which reads its status in real time. Whenever the door is opened, the ESP32 can trigger alerts, activate a buzzer, or log the event to the cloud platform. This feature enhances safety, improves monitoring accuracy, and supports preventive action in the cold chain system.

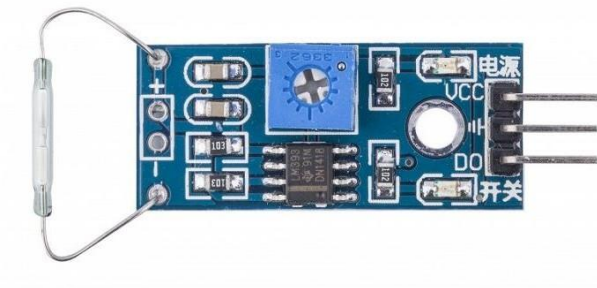


Fig.3.3.3 Magnetic Reed Sensor

Features of Sensor:

1. Simple Digital Output:

- Provides HIGH/LOW signal to the ESP32
- Easy to interface with microcontrollers

2. Low Power Consumption:

- Suitable for continuous monitoring in IoT systems

3. Reliable and Durable:

- No physical contact required for operation
- Long operational life with minimal wear

Pin Configuration:

- **VCC:** Supplies power to the module
- **GND:** Main power input
- **DATA:** Outputs the digital signal indicating door status

Working Principle of Magnetic Reed Sensor:

The magnetic reed sensor operates based on the presence of a magnetic field. It contains two thin ferromagnetic reeds enclosed inside a glass tube. When a magnet is brought close (for example, when the door is closed), the reeds come into contact, completing the circuit and indicating a “closed” state. When the door opens and the magnet moves away, the reeds separate, breaking the circuit and indicating an “open” state.

This change in state is detected by the ESP32, which then performs actions such as sending alerts, logging data, or activating warning systems. By integrating this sensor, the system can quickly respond to door-related disturbances and prevent temperature deviations.

3.3.4 LCD DISPLAY:

An **LCD (Liquid Crystal Display)** is an electronic display module widely used in embedded systems, automation, and IoT projects. It is preferred for its low power consumption, clear visibility, and ability to display alphanumeric characters and custom symbols. The most commonly used type in microcontroller-based projects is the 16x2 LCD, meaning it can display 16 characters per line across 2 lines.



Fig.3.3.4 LCD Display

Working Principle of LCD Display:

An LCD works by controlling liquid crystal molecules using an electric field to block or allow light to pass through. It has a backlight (usually LED) for illumination and polarizing filters to control visibility. The microcontroller sends electrical signals to modify the orientation of the liquid crystals, creating visible characters or graphics on the screen.

Key Features of LCD (16X2 DISPLAY):

1. **Character Display** – Can show up to 16 characters per line and 2 lines of text.
2. **Backlight Support** – Provides clear visibility in low-light conditions.
3. **Low Power Consumption** – Operates efficiently using 5V power.
4. **Parallel Interface** – Uses a 4-bit or 8-bit communication mode to connect with microcontrollers.
5. **Custom Characters** – Can store and display custom symbols.
6. **Command-based Control** – Supports various commands to clear the display, shift text, and control the cursor.

Pin Configuration of 16X2 LCD Display:

The 16x2 LCD display consists of 16 pins, each serving a unique purpose for power, control, and data transmission. It operates in 4-bit or 8-bit mode, depending on the number of data pins used.

Power Pins

The VSS pin is connected to GND (0V), while VCC is connected to a +5V power supply. The VEE pin controls the contrast of the display and is typically connected to a 10K potentiometer for adjustment.

Control Pins

The RS (register select) pin determines whether data is being sent as a command or a display character. A low (0) signal makes it a command, while a high (1) signal treats it as character data. The RW (read/write) pin selects between reading from or writing to the display, but it is usually grounded to keep the LCD in write mode. The E (enable) pin is used to trigger data transmission by sending a high-to-low pulse.

Data Pins

Pins D0 to D7 carry the actual data to be displayed. In 8-bit mode, all eight data pins (D0 to D7) are used, requiring more microcontroller pins. In 4-bit mode, only D4 to D7 are used, saving I/O pins while still enabling full functionality.

Backlight Pins

The LCD has two additional pins for the backlight. The LED+ pin is connected to +5V, while LED- is connected to GND. This powers the backlight, making the display visible in low-light conditions.

Interfacing with Arduino

In a typical 4-bit mode connection, RS, E, D4, D5, D6, and D7 are connected to digital pins of an Arduino. The RW pin is grounded, and a potentiometer is used to adjust contrast via the VEE pin. The LED+ pin is powered by 5V, and LED- is connected to GND for backlight operation.

3.3.5 LED INDICATOR:

The LED indicator is used to provide visual status feedback of the cold storage system. It helps in quickly identifying system conditions such as normal operation, alerts, or faults without the need for a display interface. LEDs are simple, reliable, and ideal for real-time indication in embedded systems.

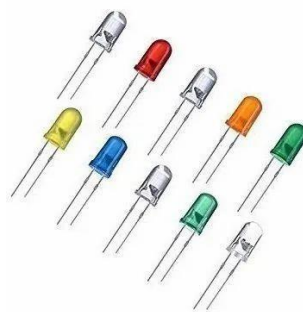


Fig.3.3.5 LED Indicator

Features of LED Indicator:

1. Visual Status Indication:

- Provides immediate visual feedback of system conditions
- Useful for monitoring without accessing the dashboard

2. Fast Response

- Instantly reflects changes in system status
- Enables quick decision-making

3. Simple Interfacing:

- Easily connected to ESP32 GPIO pins
- Controlled using basic digital signals

Working Principle of LED Indicator:

An LED (Light Emitting Diode) emits light when current flows through it in the forward direction. The ESP32 controls the LED by sending HIGH or LOW signals to the connected GPIO pins. When the output pin is HIGH, current flows and the LED glows; when LOW, the LED remains off. Based on sensor data from the system, the microcontroller activates specific LEDs to indicate different environmental conditions, enabling real-time visual monitoring.

3.3.6 BUZZER:

The buzzer provides an audible alert when abnormal conditions such as temperature deviation or door opening are detected.



Fig.3.3.6 Buzzer

Features of Buzzer:

1. Audible Alert System:

- Produces sound to indicate abnormal conditions
- Useful in environments where visual monitoring is limited

2. Instant Notification

- Provides real-time alerts for temperature or humidity deviations
- Ensures quick response to prevent damage to stored goods

3. Low Power Consumption:

- Consumes minimal power, suitable for continuous monitoring systems
- Efficient for IoT-based applications

Working Principle of Buzzer:

A buzzer converts electrical signals into sound waves. When the ESP32 sends a HIGH signal to the buzzer, it activates and produces sound. In the case of an active buzzer, the internal circuit generates the tone automatically. For a passive buzzer, the ESP32 generates pulses (PWM signals) to create different sound frequencies. The buzzer is triggered based on sensor data, such as when temperature or humidity crosses predefined thresholds, providing an immediate audible warning for critical conditions.

3.3.7 Relay Module:

The relay module acts as a switching device to control the cooling system. It is triggered by the ESP32 based on sensor readings.



Fig.3.3.7 Relay Module

Features of Relay Module:

1. High Voltage Control:

- Allows control of AC/DC appliances using low-voltage signals
- Suitable for switching cooling systems and other heavy loads

2. Digital Control Interface:

- Operated using simple HIGH/LOW signals from ESP32
- Easy integration with embedded systems

3. Multiple Channel Options:

- Available in single or multi-channel configurations
- Enables control of multiple devices simultaneously

Working Principle of Relay Module:

A relay module works as an electrically operated switch. When the ESP32 sends a signal to the relay input, it energizes an internal coil, creating a magnetic field that moves a mechanical contact. This action either opens or closes the circuit connected to the external device. In a cold storage system, the relay is activated when temperature or humidity exceeds set limits, allowing the microcontroller to turn ON or OFF devices like compressors or fans automatically, maintaining optimal storage conditions.

3.3.8 PELTIER MODULE WITH COOLING FAN AND HEAT SINK

The relay module is used to control high-power electrical devices such as compressors, fans, or cooling systems in the cold storage setup. It acts as a switching interface between the low-power ESP32 and high-voltage equipment, ensuring safe and efficient operation.



Fig.3.3.8 Peltier Module with cooling fan and heat sink

Features of Peltier Module:

1. Solid-State Cooling:

- No moving parts, ensuring silent operation
- High reliability with minimal maintenance

2. Dual Functionality:

- Can be used for both cooling and heating by reversing polarity
- Provides flexibility in temperature control

3. Precise Temperature Control:

- Enables accurate regulation when combined with sensors and controllers
- Suitable for sensitive storage environments

Working Principle of Peltier Module:

The Peltier module operates based on the **thermoelectric effect**, specifically the Peltier effect. When direct current passes through the module, heat is absorbed on one side (creating a cooling effect) and released on the opposite side (creating a heating effect). The cold side is placed inside the storage chamber, while the hot side is connected to a heat sink and fan to dissipate heat. The ESP32 controls the module based on temperature readings, turning it ON or OFF to maintain the desired cooling conditions.

3.3.9 SIM900A GSM MODULE:

The GSM module is primarily used to transmit alert messages in the form of SMS when abnormal conditions, such as temperature deviations or system failures, are detected. The module operates using a SIM card and communicates with the ESP32 microcontroller through serial communication (UART).



Fig.3.3.9 SIM900A GSM Module

Features of GSM Module:

1. Wireless Communication:

- Enables data transmission over cellular networks
- Suitable for remote locations without Wi-Fi access

2. SMS Alert Capability:

- Sends instant text messages for critical conditions
- Ensures users are notified of temperature or humidity changes

3. Low Power Operation:

- Designed for efficient energy usage
- Suitable for IoT-based continuous monitoring systems

Working Principle of GSM Module:

The SIM900A GSM module operates by connecting to a cellular network using a SIM card. It communicates with the ESP32 through serial communication (UART). When the microcontroller detects abnormal conditions from sensors, it sends commands to the GSM module to transmit SMS alerts to predefined phone numbers. The module converts these commands into signals that are sent over the mobile network, ensuring real-time remote notification and system awareness.

3.3.10 LM2596 BUCK CONVERTER

The LM2596 module steps down the 12V input to a regulated 5V output, which powers components such as ESP32, LCD, relay, and GSM module.

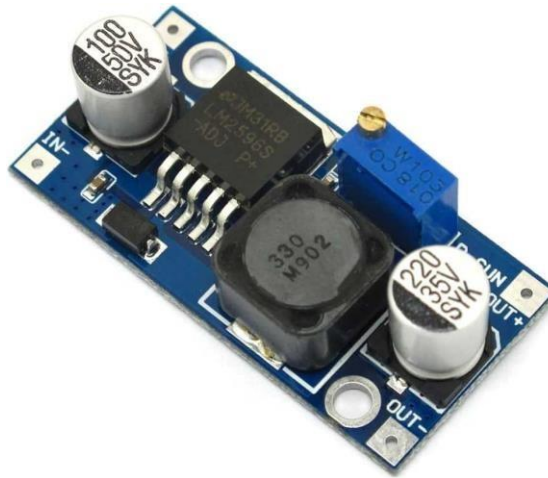


Fig.3.3.10 LM2596 Buck Converter

Features of GSM Module:

1. Step-Down Voltage Conversion:

- Converts higher DC voltage to lower DC voltage efficiently
- Suitable for powering low-voltage components from higher sources

2. Compact and Reliable:

- Small size, easy to integrate into circuits
- Suitable for continuous operation in embedded systems

3. High Efficiency:

- Offers efficiency up to 90% or more
- Minimizes energy loss in the form of heat

Working Principle of Buck Converter:

The LM2596 operates based on the principle of a switching regulator. It rapidly switches the input voltage ON and OFF using an internal transistor, controlling the energy transferred to the output through an inductor, diode, and capacitor. By adjusting the duty cycle of the switching signal, the converter regulates the output voltage to a lower, stable level. This stepped-down voltage is then supplied to components like the ESP32, ensuring efficient and safe operation of the overall system.

3.3.11 BREADBOARD

The breadboard is used for prototyping and testing electronic circuits without the need for soldering. It allows easy assembly, modification, and debugging of the cold storage system during the development phase, making it ideal for experimentation and design validation

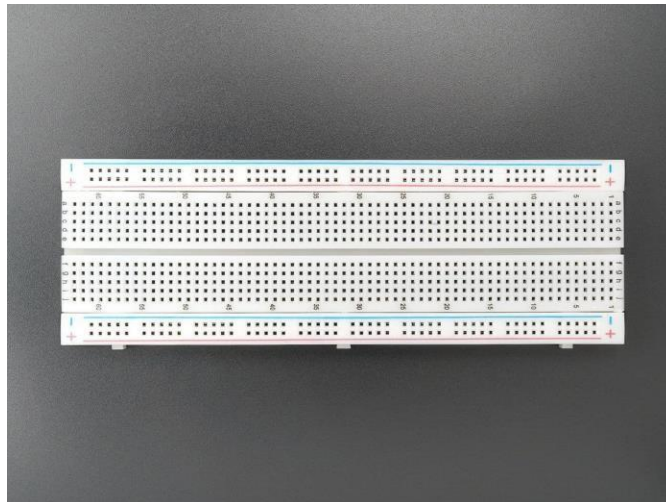


Fig.3.3.11 Breadboard

Features of Breadboard:

1. Solderless Design:

- Enables circuit construction without permanent connections
- Allows easy insertion and removal of components

2. Reusable Platform:

- Can be used multiple times for different circuit designs
- Ideal for iterative testing and development

3. Easy Component Integration:

- Supports a wide range of electronic components like resistors, LEDs, and ICs
- Facilitates quick prototyping of embedded systems

Working Principle of Breadboard:

A breadboard consists of a grid of holes internally connected by metal strips. When components and wires are inserted into these holes, electrical connections are established through these conductive paths. The central area is typically used for placing components, while the side rails are used for power supply distribution. This arrangement allows the ESP32 and other components to be interconnected easily, enabling quick testing and modification of the circuit without permanent soldering.

CHAPTER 4

SOFTWARE USED

4.1 ARDUINO IDE

Arduino IDE is the primary software tool used for developing and managing the embedded program in the cold storage monitoring system. It provides an integrated environment where code can be written, compiled, and uploaded to the ESP32 microcontroller. The platform simplifies the interaction between hardware components and software logic, making it suitable for rapid prototyping and system development.

Key Features:

1. Simple Programming Environment:

- Provides a clean interface for writing and editing code
- Supports C/C++ based embedded programming

2. Compilation and Uploading:

- Converts written code into machine-level instructions
- Allows direct uploading to ESP32 via USB interface

3. Library Integration:

- Supports built-in and external libraries
- Enables easy interfacing with sensors like DHT22 and modules like GSM

4. Serial Communication Support:

- Includes a serial monitor for real-time data visualization
- Helps in debugging and testing system behavior

5. Board Configuration:

- Supports multiple microcontroller boards including ESP32
- Allows selection of ports, baud rate, and board setting

Application in the Project:

In this project, Arduino IDE is used to design and implement the complete control logic of the system. It reads data from sensors such as temperature and humidity sensors, processes the values, and compares them with predefined thresholds. Based on this analysis, it controls output devices like relays, buzzers, and cooling modules. Additionally, it manages communication with external modules such as GSM for sending alerts. The software plays a central role in coordinating all system operations.

Advantages:

- Easy to learn and use for embedded system development
- Enables quick coding, testing, and modification
- Supports a wide range of hardware components
- Large community support and documentation availability
- Suitable for rapid prototyping and academic projects

Working Principle of Arduino IDE:

The Arduino IDE works by translating the user-written program into a format that the microcontroller can understand. Once the code is compiled, it is uploaded to the ESP32 through a communication interface. The microcontroller then executes the instructions continuously in a loop, reading sensor inputs, processing data, and controlling connected devices accordingly. The serial monitor helps in observing outputs and diagnosing errors during execution.

4.2 ANDROID STUDIO

Android Studio is used for developing the mobile application that enables users to monitor and control the cold storage system remotely. It provides a comprehensive environment for designing user interfaces, integrating backend services, and implementing real-time data interaction between the system and the user.

Key Features:

1. Integrated Development Environment:

- Provides tools for coding, designing, and testing applications
- Supports development using Java and Kotlin

2. User Interface Design Tools:

- Offers drag-and-drop layout editor
- Helps in creating interactive and responsive app interfaces

3. Real-Time Data Integration:

- Enables connection with cloud platforms or communication protocols
- Supports real-time data display and updates

4. Emulator Support:

- Allows testing of applications without a physical device
- Simulates different screen sizes and Android versions

5. API and Library Support::

- Supports integration of external libraries and APIs
- Simplifies implementation of advanced features

Application in the Project:

In this project, Android Studio is used to develop a mobile application that displays temperature and humidity data received from the system. It allows users to monitor real-time conditions and receive alerts when values exceed safe limits. The application acts as a user interface for remote interaction, improving accessibility and control over the cold storage environment.

Advantages:

- Enables development of user-friendly mobile applications
- Supports real-time monitoring and control
- Provides powerful tools for testing and debugging
- Flexible and scalable for future enhancements

Working Principle of Android Studio:

Android Studio functions by allowing developers to write application code and design user interfaces within a single platform. The code is compiled into an Android application package (APK), which can be installed on mobile devices. The application communicates with the system through defined protocols or cloud services, receives data, and updates the interface accordingly. This enables users to monitor system conditions and receive notifications in real time.

4.3 MOBILE APPLICATION

The cold chain monitoring mobile application is developed to display real-time temperature and humidity values from the storage unit, enabling users to monitor environmental conditions remotely. The application communicates with the ESP32-based system through IoT integration and receives continuous sensor data for live tracking.

When the user launches the application, a dashboard interface is displayed showing system parameters and connection status. The user can initiate communication with the system through a connection option provided in the app. Once the connection is successfully established, the ESP32 begins transmitting real-time data, including temperature and humidity readings, to the application.

The received data is updated dynamically on the dashboard, allowing users to continuously observe storage conditions. In case of abnormal variations beyond the defined threshold limits, the system triggers alerts, ensuring timely action to prevent damage to temperature-sensitive products.

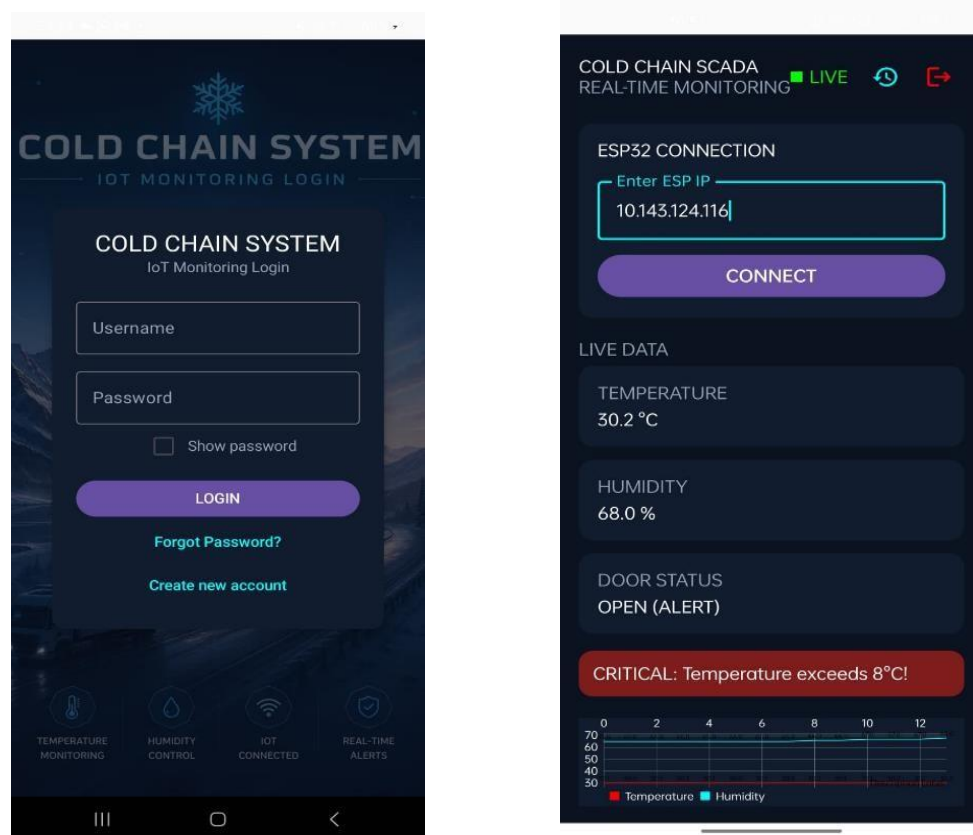


Fig.4.3.1 Cold Chain System Mobile Application

The mobile application displays the temperature value clearly at the top of the screen, allowing users to quickly view the current condition of the cold storage system. The interface is designed to present data in a simple and readable format for effective monitoring..

Below the main display, the application shows:

- Temperature (in °C)
- Humidity (in %)
- System Status (Normal / Alert condition)

The data is updated continuously as it is received from the ESP32, ensuring real-time monitoring without requiring the user to be physically present near the storage unit. This allows users to track environmental conditions efficiently and respond quickly to any abnormal changes, helping to maintain optimal storage conditions and prevent spoilage.

4.4 WEB DASHBOARD

The web dashboard is developed as a centralized monitoring interface for the cold chain system. It enables users to access and view system data through a web browser, providing a convenient platform for observing environmental conditions and system performance.

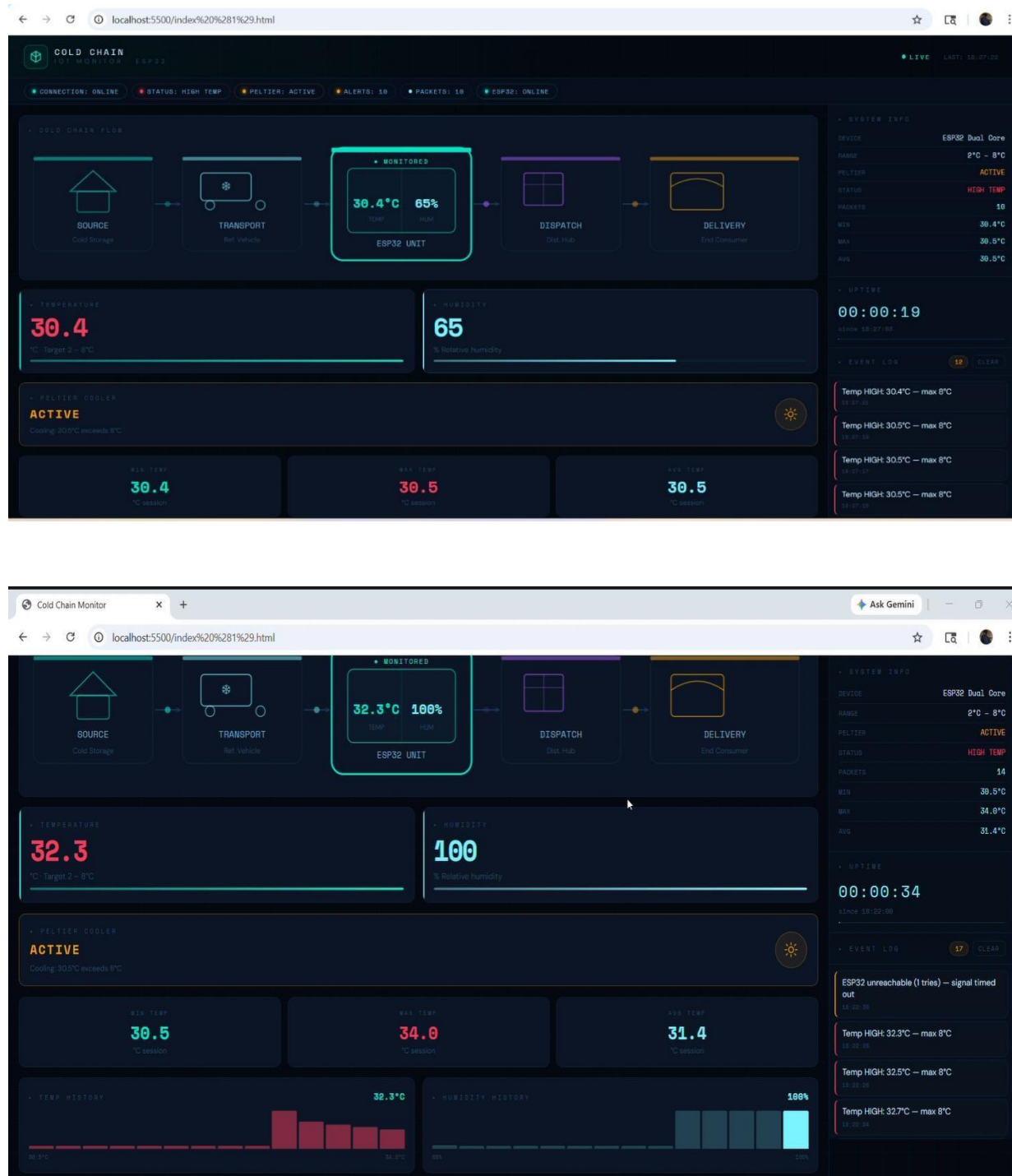


Fig.4.4.1 Cold Chain System Web Dashboard

The dashboard displays key parameters such as temperature (in °C), humidity (in %), and the operational status of the cooling system. It presents real-time data received from the ESP32 in a clear and organized format, allowing users to easily understand whether the storage environment is maintained within the required threshold range of 2°C to 8°C.

The data is updated dynamically as new readings are transmitted, ensuring accurate and continuous monitoring. With its simple and user-friendly design, the dashboard can be accessed from any location, enabling users to track system behavior effectively and stay informed about changes in storage conditions.

CHAPTER 5

WORKING

5.1 WORKING MODEL

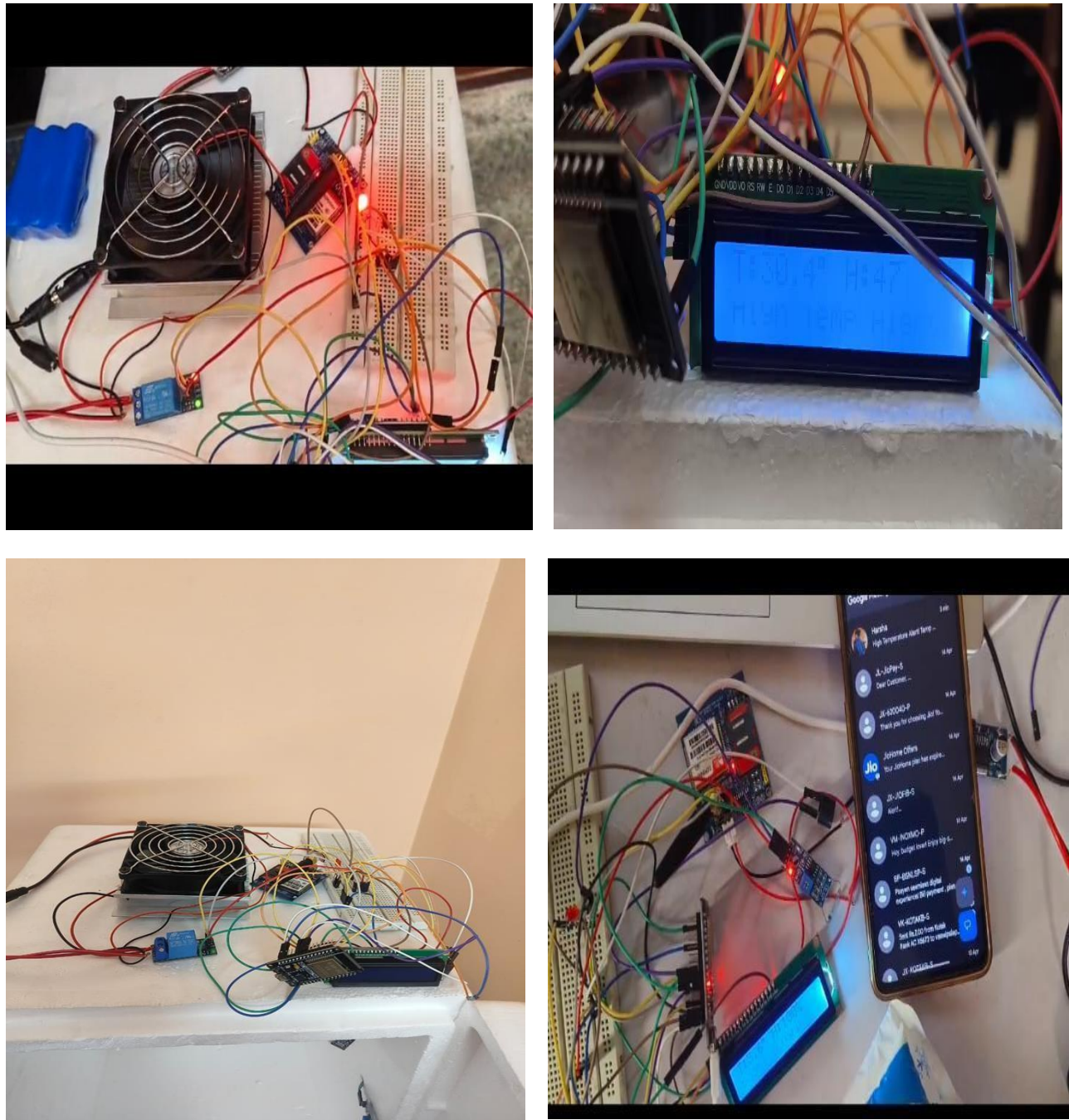


Fig.5.1.1 Working Model of IoT Driven Cold Chain System

1. System Initialization:

The operation of the system begins with the initialization phase, where power is supplied to the ESP32 microcontroller and all connected components. These include sensors, LCD display, GSM module, and output devices. After initialization, the ESP32 establishes a Wi-Fi connection for communication with the mobile application. Simultaneously, the GSM module is configured to enable SMS-based alert notifications.

2. Data Acquisition (Sensing Layer):

The sensing layer consists of a DHT22 sensor and a magnetic reed switch module. The DHT22 sensor continuously measures temperature and humidity inside the cold storage unit, while the reed switch module detects the door status (open or closed). These sensors provide real-time environmental data to the ESP32 microcontroller..

3. Data Processing and Decision Making:

The ESP32 microcontroller processes the received sensor data and compares the temperature values with predefined threshold limits, typically maintained within the range of 2°C to 8°C. Based on this comparison, the system classifies the operating condition as normal, warning, or critical. This decision-making process forms the core logic of the system.

4. Local Monitoring Using LCD Display:

For on-site monitoring, the processed data is displayed on an LCD screen. The display shows real-time values of temperature, humidity, and door status. This ensures that users can monitor system conditions locally even without internet connectivity.

5. Remote Monitoring Through Mobile Application:

The ESP32 transmits real-time data to a mobile application via Wi-Fi. The application provides a user interface for monitoring system parameters, including temperature, humidity, and door status. It also displays graphical trends and system status indicators, enabling effective remote supervision.

6. Alert and Notification System:

When abnormal conditions are detected, such as temperature exceeding the safe range or the door remaining open, the system activates the alert mechanism. Local alerts include

buzzer activation and LED indication. Additionally, the GSM module sends SMS notifications to the user, ensuring reliable alerts even in the absence of internet connectivity.

7. Cooling System Operation:

To maintain the required temperature conditions, the system includes an automated cooling mechanism. When the temperature rises above the threshold value, the ESP32 activates a relay or MOSFET driver to power the Peltier module and cooling fan. The cooling system reduces the temperature and restores it to the acceptable range. Once the temperature stabilizes, the system automatically deactivates the cooling components.

8. Continuous Monitoring and Control:

The system operates continuously in a loop where sensing, data processing, decision-making, communication, and control actions are performed repeatedly. This closed-loop operation ensures real-time monitoring, quick response to environmental changes, and efficient maintenance of optimal cold chain condition.

CHAPTER 6

RESULT

6.1 INTRODUCTION

The IoT-based Cold Chain Monitoring System was tested under different operating conditions to evaluate its performance, accuracy, and reliability. The results obtained from LCD display, GSM alerts, mobile application, and website dashboard are discussed in the following subsections.

6.2 LCD DISPLAY RESULTS

The LCD display provided real-time monitoring of temperature, humidity, and door status. The values were updated continuously without delay, ensuring reliable local monitoring. The display clearly indicated normal, warning, and critical conditions based on threshold limits.

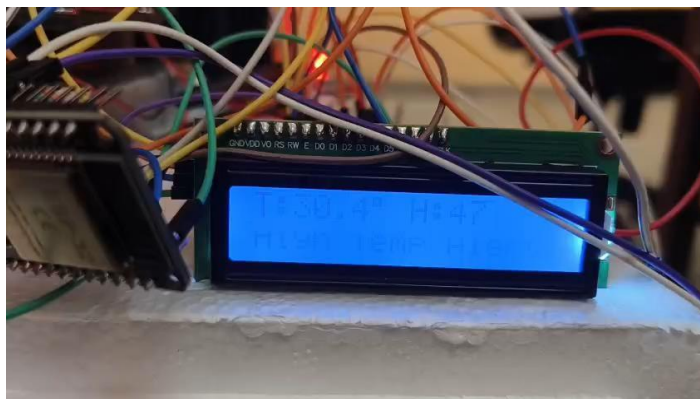


Fig.6.2.1 LCD Display Result

6.3 MOBILE APPLICATION RESULTS

The mobile application successfully displayed real-time data received from the ESP32 via Wi-Fi. It provided live temperature and humidity values, door status, and graphical trends. Warning and critical alerts were generated accurately. The application enabled remote monitoring and ensured quick response to abnormal conditions.



Fig.6.3.1 Mobile Application Result

6.4 WEBSITE DASHBOARD RESULTS

The website dashboard displayed real-time data updates similar to the mobile application. It provided a centralized interface for monitoring system parameters and historical trends. The dashboard ensured continuous visibility of system performance and enhanced data accessibility.

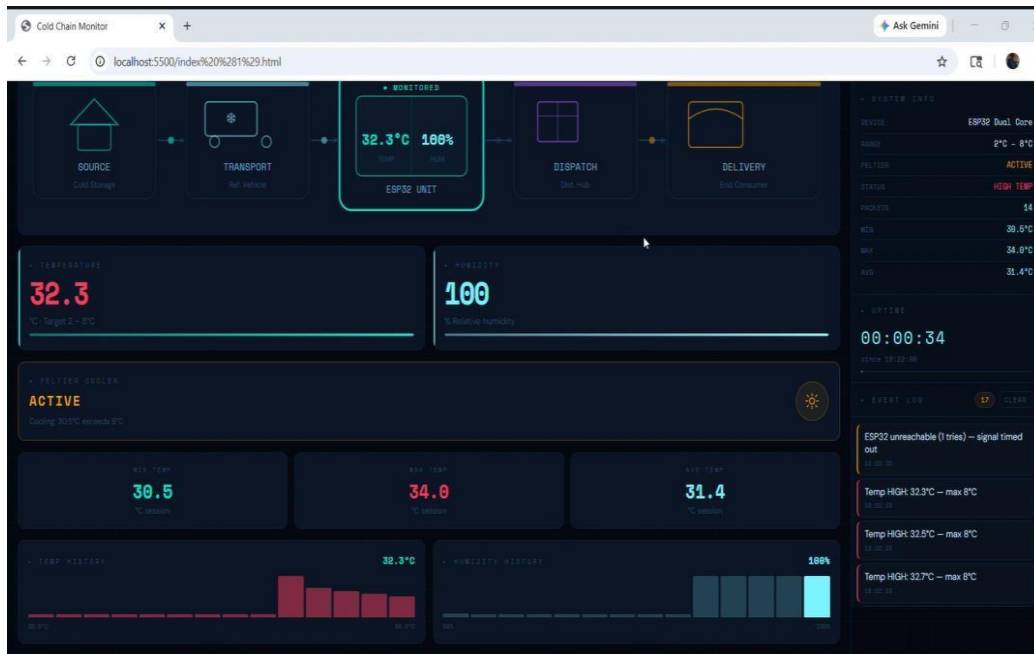


Fig.6.4.1 Web Dashboard Result

6.5 GSM ALERT RESULTS

The GSM module successfully transmitted SMS alerts during abnormal conditions such as temperature exceeding limits or door opening. The alerts were received instantly on the registered mobile number. This ensured reliable communication even in the absence of internet connectivity.



Fig.6.5.1 GSM Alert Result

6.6 PERFORMANCE ANALYSIS

The system demonstrated accurate sensing, fast response time, and reliable communication. The integration of LCD, mobile application, website dashboard, and GSM module ensured efficient monitoring and control. Overall, the system maintained temperature within the required range and responded effectively to all tested conditions.

1. Accurate Monitoring of Storage Conditions:

The DHT22 sensor continuously measured temperature and humidity inside the cold storage unit with good accuracy. This ensured that the environmental conditions remained within the required range for storing temperature-sensitive products.

2. Successful Real-Time Display and Remote Monitoring:

The system displayed live temperature and humidity values on both the LCD and mobile application. The ESP32 transmitted data efficiently, allowing users to monitor storage conditions remotely without being physically present..

3. Automated Temperature Control Mechanism::

The cooling system, controlled through the relay and Peltier module, responded automatically to changes in temperature. This helped maintain stable internal conditions and reduced dependency on manual intervention.

4. Effective Humidity Management::

The system continuously tracked humidity levels and responded when values exceeded safe limits. This prevented issues such as moisture buildup, condensation, and potential spoilage of stored products.

5. Reliable Alert and Notification System:

The buzzer and GSM module worked effectively to alert users during abnormal conditions. Immediate notifications enabled quick corrective action, improving the safety and reliability of the system.

6. Continuous Real-Time Data Monitoring:

The mobile application consistently displayed updated temperature and humidity readings received from the ESP32. This ensured uninterrupted monitoring and allowed users to track system performance at all times.

CHAPTER 7

SUMMARY

7.1 ADVANTAGES:

- **Real-Time Monitoring:**– The system continuously tracks temperature and humidity, ensuring that storage conditions are always visible and under control.
- **Improved Product Safety:**– By maintaining optimal environmental conditions, the system helps preserve the quality of temperature-sensitive products such as vaccines and perishable goods.
- **Automated Control:**– The integration of sensors with relay and cooling modules allows automatic regulation of temperature, reducing the need for manual intervention.
- **Instant Alert System:**– The buzzer and GSM module provide immediate notifications when conditions exceed safe limits, enabling quick response to prevent damage or spoilage.
- **Remote Accessibility:**– Users can monitor system parameters through a mobile application, allowing supervision from any location without physical presence.
- **Reliable Communication:**– The system supports both IoT-based monitoring and GSM alerts, ensuring communication even in limited internet conditions.
- **Low Power Consumption:**– Efficient components like ESP32 and optimized modules ensure reduced energy usage, supporting continuous operation.

7.2 DISADVANTAGES:

- **Initial Setup Cost** – The integration of sensors, communication modules, and control systems increases the initial investment compared to basic storage methods.
- **System Complexity:**– Integration of multiple hardware and software components increases system complexity and requires proper configuration
- **Maintenance Requirement:**– Sensors and electronic modules require periodic calibration and maintenance to ensure accurate performance.
- **Limited Cooling Capacity:**– The Peltier module is suitable for small-scale applications and may not be efficient for large storage systems.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION:

The IoT-based Cold Chain Monitoring System developed in this project offers a reliable and efficient solution for maintaining temperature-sensitive environments with continuous real-time supervision. The system combines sensing, processing, communication, and control mechanisms to ensure that storage conditions remain within the required limits at all times.

The DHT22 sensor was used to measure temperature and humidity with consistent accuracy, while the magnetic reed switch module enabled effective detection of door status. The ESP32 microcontroller functioned as the core processing unit, continuously collecting sensor data and comparing it with predefined threshold values. Based on this evaluation, the system initiated appropriate control actions, such as activating the cooling mechanism whenever the temperature exceeded the safe range of 2°C to 8°C.

For local monitoring, the LCD display provided real-time updates of environmental parameters, allowing users to observe system status directly at the storage location. In addition, the mobile application enabled remote monitoring by presenting live temperature, humidity, and door status data, ensuring accessibility from any location.

To enhance system reliability, the GSM module was incorporated to deliver SMS-based alerts during critical situations. Whenever abnormal conditions such as temperature deviation or extended door opening were detected, the system generated immediate notifications, allowing timely corrective action.

The cooling system, implemented using a Peltier module along with a fan, responded dynamically to environmental changes and maintained stable temperature conditions. The inclusion of buzzer and LED indicators further strengthened the alert mechanism by providing instant local warnings. The system was evaluated under different operating conditions and demonstrated stable performance, accurate sensing, and reliable communication. Overall, the proposed solution is cost-effective, adaptable, and suitable for applications in healthcare, food storage, and pharmaceutical cold chain management.

8.2 FUTURE SCOPE:

8.2.1 Location-Aware Cold Chain Tracking:

Integration of a GPS module to monitor the real-time location of storage units during transportation, improving traceability and logistics management.

8.2.2 Predictive Failure Detection:

Application of Artificial Intelligence and Machine Learning techniques to analyze sensor trends and predict potential failures before they occur.

8.2.3 Multi-Parameter Monitoring:

Addition of advanced sensors to track parameters such as air quality, gas leakage, and pressure for comprehensive environmental monitoring.

8.2.4 Scalability for Industrial Applications:

Expansion of the system to support large-scale cold chain networks with multiple interconnected storage units.

8.2.5 Security Enhancement::

Incorporation of authentication mechanisms and data encryption to ensure secure communication and prevent unauthorized access.

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